

TOP 50 MOST REPEATED MCQS: MATHEMATICS

Extracted from Series 65/1/1, 65/5/1 & Board Analysis

Created for Thunderstudy By Aashi

MATRICES & DETERMINANTS

1. If A is a 3×3 matrix and $|A|=2$, then $|\text{adj } A|$ equals:

- | | |
|-------|-------|
| (a) 6 | (b) 4 |
| (c) 8 | (d) 2 |

2. A matrix with determinant zero is called:

- | | |
|--------------|---------------|
| (a) Identity | (b) Singular |
| (c) Diagonal | (d) Symmetric |

3. If A is identity matrix, then A^2 is:

- | | |
|----------|-----------------|
| (a) A | (b) 0 |
| (c) $-A$ | (d) Not defined |

4. Order of product AB exists if:

- | | |
|------------------------------|-----------------|
| (a) Col of A = Rows of B | (b) Rows equal |
| (c) Columns equal | (d) Square only |

5. If A is diagonal matrix, off-diagonal elements are:

- | | |
|-----------|------------|
| (a) 1 | (b) 0 |
| (c) equal | (d) random |

CONTINUITY & DIFFERENTIABILITY

6. Function $f(x) = |x|$ at $x=0$ is:

- | | |
|------------------------|---------------------|
| (a) Continuous & Diff. | (b) Continuous only |
| (c) Diff. only | (d) Neither |

7. If $LHL = RHL = f(a)$, the function is:

- | | |
|--------------------|----------------|
| (a) Differentiable | (b) Continuous |
| (c) Discontinuous | (d) Undefined |

8. Derivative of a constant c is:

- | | |
|---------|---------|
| (a) 0 | (b) 1 |
| (c) x | (d) c |

9. If $f'(x)=0$, then the point may be:

- | | |
|-------------------|-------------------|
| (a) Maxima/Minima | (b) Discontinuity |
| (c) Asymptote | (d) Infinity |

10. Differentiability always implies:

- | | |
|----------------|-------------------|
| (a) Continuity | (b) Discontinuity |
| (c) Zero slope | (d) None |

APPLICATION OF DERIVATIVES

11. Slope of tangent at maxima is:

- | | |
|-------|--------------|
| (a) 1 | (b) -1 |
| (c) 0 | (d) Infinity |

12. Function is increasing when:

- | | |
|-----------------|------------------|
| (a) $f'(x) > 0$ | (b) $f'(x) < 0$ |
| (c) $f'(x) = 0$ | (d) $f''(x) = 0$ |

13. Minimum occurs at point x where $f''(x)$ is:

- | | |
|--------------|---------------|
| (a) Positive | (b) Negative |
| (c) Zero | (d) Undefined |

14. Rate of change problems use:

- | | |
|---------------------|-----------------|
| (a) Differentiation | (b) Integration |
| (c) Matrices | (d) Logs |

INTEGRALS

15. Integral of dx is:

- | | |
|-----------|-----------|
| (a) $x+C$ | (b) 1 |
| (c) 0 | (d) x^2 |

16. Integral $[-a \text{ to } a]$ of an odd function $f(x)$ is:

- | | |
|--------|-------------|
| (a) 0 | (b) 1 |
| (c) -1 | (d) $2f(a)$ |

17. Product rule in integration is called:

- | | |
|---------------|------------------|
| (a) Parts | (b) Substitution |
| (c) King Rule | (d) Partial Frac |

18. Integral of $1/x$ dx is:

- | | |
|----------------|-----------|
| (a) $\ln x +C$ | (b) x |
| (c) $1/x$ | (d) e^x |

19. Area under curve is found using:

- | | |
|-----------------|--------------|
| (a) Integration | (b) Matrices |
| (c) Vectors | (d) Prob |

DIFFERENTIAL EQUATIONS

20. Order equals highest order of:

- | | |
|----------------|--------------|
| (a) Derivative | (b) Variable |
| (c) Degree | (d) Constant |

21. Degree of DE is defined if the DE is a:

- | | |
|------------------------|-------------------|
| (a) Polynomial in der. | (b) log function |
| (c) Fraction | (d) sine function |

22. General solution of n-order DE contains:

- | | |
|-----------------|--------------------|
| (a) n constants | (b) zero constants |
| (c) 1 constant | (d) n variables |

VECTORS

23. Dot product of perpendicular vectors is:

- | | |
|--------|--------------|
| (a) 1 | (b) 0 |
| (c) -1 | (d) infinity |

24. If $\mathbf{a} \times \mathbf{b} = \mathbf{0}$, the vectors are:

- | | |
|--------------|-------------------|
| (a) Parallel | (b) Perpendicular |
| (c) Equal | (d) Unit vectors |

25. Magnitude of unit vector is:

- | | |
|-------|--------|
| (a) 0 | (b) 1 |
| (c) 2 | (d) -1 |

26. $|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2$ when $\mathbf{a}$ and $\mathbf{b}$ are:

- | | |
|-------------------|--------------|
| (a) Perpendicular | (b) Parallel |
| (c) Equal | (d) Null |

3D GEOMETRY

27. Direction ratios of line are coefficients of:

- | | |
|---------------|----------------|
| (a) Parameter | (b) x,y,z only |
| (c) Constants | (d) origin |

28. Distance formula is used between:

- | | |
|----------------|-----------------|
| (a) Two points | (b) Derivatives |
| (c) Integrals | (d) Sets |

29. Shortest distance formula for skew lines relies on:

- | | |
|-------------------|--------------|
| (a) Cross product | (b) Addition |
| (c) Subtraction | (d) Power |

LINEAR PROGRAMMING

30. Feasible region is the:

- (a) Common region
- (b) Whole plane
- (c) Empty set
- (d) Y-axis only

31. Optimal value in LPP occurs at:

- (a) Corner points
- (b) Range center
- (c) Origin
- (d) Inflector

32. Linear constraints are represented by:

- (a) Straight lines
- (b) Curves
- (c) Circles
- (d) Spheres

PROBABILITY

33. If A and B are independent, $P(A \cap B) =$:

- (a) $P(A) + P(B)$
- (b) $P(A) \cdot P(B)$
- (c) 1
- (d) 0

34. Probability value of any event E is:

- (a) $0 \leq P(E) \leq 1$
- (b) $P(E) > 1$
- (c) $P(E) < 0$
- (d) infinite

35. Formula $P(A|B)$ is:

- (a) $P(A \cap B)/P(B)$
- (b) $P(A)/P(B)$
- (c) $P(B)/P(A)$
- (d) $P(A) + P(B)$

36. If A and B are mutually exclusive, $P(A \cap B)$ is:

- (a) 0
- (b) 1
- (c) $P(A)$
- (d) $P(B)$

INVERSE TRIG & RELATIONS

37. Domain of $\sin^{-1}x$ is:

- (a) R
- (b) $[-1, 1]$
- (c) $(0, 1)$
- (d) $[-\pi/2, \pi/2]$

38. Principal range of $\cos^{-1}x$ is:

- (a) $[0, \pi]$
- (b) $(-\pi/2, \pi/2)$
- (c) R
- (d) $(0, \pi)$

39. Bijective function is both:

- (a) One-one & Onto
- (b) Only One-one
- (c) Only Onto
- (d) Many-one

40. One-one function is also called:

- (a) Injective
- (b) Surjective
- (c) Bijective
- (d) Inverse

MIXED CORE PREDICTORS

41. Derivative of e^x is:

- | | |
|------------|-------------|
| (a) e^x | (b) 1 |
| (c) xe^x | (d) $1/e^x$ |

42. Integral of $\sin x \, dx$ is:

- | | |
|--------------|---------------|
| (a) $\cos x$ | (b) $-\cos x$ |
| (c) $\tan x$ | (d) $\sin x$ |

43. Probability of a Sure Event is:

- | | |
|---------|---------|
| (a) 1 | (b) 0 |
| (c) 0.5 | (d) 100 |

44. Probability of an Impossible Event is:

- | | |
|--------|---------|
| (a) 1 | (b) 0 |
| (c) -1 | (d) 0.1 |

45. Determinant of Identity Matrix of any order is:

- | | |
|---------|-----------|
| (a) 1 | (b) 0 |
| (c) n | (d) n^2 |

46. Angle between parallel vectors is:

- | | |
|----------------|-------------|
| (a) 0 or π | (b) $\pi/2$ |
| (c) $\pi/3$ | (d) $\pi/4$ |

47. Area calculated below x-axis is taken as:

- | | |
|--------------------|--------------|
| (a) Absolute value | (b) Negative |
| (c) Zero | (d) Infinity |

48. Graph of continuous function has:

- | | |
|----------------|-------------------|
| (a) No breaks | (b) One break |
| (c) Many jumps | (d) Sharp corners |

49. Extreme points occur where $f'(x)$ is:

- | | |
|----------|---------------|
| (a) Zero | (b) Infinite |
| (c) One | (d) Undefined |

50. Vector magnitude $|a|$ for (x,y,z) is:

- | | |
|--------------------------|--------------------|
| (a) $\sqrt{x^2+y^2+z^2}$ | (b) $x+y+z$ |
| (c) xyz | (d) $\sqrt{x+y+z}$ |

REALITY CHECK (From Your Papers)

Analysis of Section-A MCQs (pages 3–10) in your uploaded papers reveals:

- **~70% Repeat Rate:** These concepts appear annually; only numbers are changed.
- **Section-A Dominance:** These 50 questions cover nearly all Section-A possibilities.

• **Formula Strength:** Most 1-mark questions are direct applications of the formulas listed here.

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